How EdgeValue builds a message

Add learned pair information to the particle value, then apply the attention weight.

1. Encode the pair

Pair physics features

Separation, momentum sharing, pair mass, ...

Learned pair mapping

e_{ij}

Relationship vector

2. Add to the value

Information about particle j

v_j

$+$

Information about the pair

e_{ij}

$v_j + e_{ij}$

Message content

3. Weight and send

Apply attention weight a_{ij}

$$a_{ij}\,(v_j + e_{ij})$$

Receiver i

Sums these contributions from all sender particles